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Fitness flux in SARS-CoV-2 and influenza H3N2

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eLife Assessment

This **valuable** study presents a comparative framework for quantifying adaptation in SARS-CoV-2 and influenza H3N2 from changes in clade frequencies, enabling comparison of rates of adaptation across the two viruses on a common per-generation scale and showing that a simple count of mutations performs as well as much more sophisticated protein language models in predicting variant fitness. The evidence supporting the quantitative estimates of fitness flux is currently **incomplete** because the analysis assumes a single, time-independent fitness for each clade; for antigenically evolving viruses, clade fitness changes with population immunity, and it remains unclear how this assumption affects the estimated fitness flux. The interpretation of cumulative fitness flux in relation to changes in absolute fitness over time also requires clarification.

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Abstract

The tempo of viral adaptation is usually read indirectly from the composition of mutations, through measures such as dN/dS. Here we measure it directly from the dynamics of variant frequencies, where we use multinomial logistic regression to estimate a fitness for each co-circulating variant. We aggregate these estimates to derive the rate of change of mean population fitness, referred to as fitness flux. Tracing SARS-CoV-2 from its emergence, we find that it initially adapted rapidly, doubling in fitness every 6 months from Jan 2021 to Jun 2022, but slowing to every 2.4 years from Jul 2022 to Dec 2025. Seasonal influenza H3N2 sustained a slower, steadier pace doubling in fitness every 10.0 years. In both, the rate of fitness gain closely tracks the variance in fitness, matching the 1:1 expectation of Fisher's fundamental theorem. Phylogenetic contrasts between parent and child lineages localize most fitness gain to spike, and within spike to the receptor-binding domain, where a simple count of spike S1 substitutions predicts lineage fitness about as well as deep-learning escape and protein-language-model scores. Measuring fitness directly thus offers a transparent, frequency-based alternative to mutational proxies for tracking and anticipating viral adaptation. The website *blab.github.io/fitness-flux/* is the intended reading experience of this paper, providing responsive layout and interactive figures.

Introduction

RNA viruses evolve rapidly and the selective pressures they face shift over the course of emergence to endemicity. A newly established virus initially adapts to its new host by refining its capacity for within-host replication and between-host transmission. Once a virus becomes endemic, adaptation is instead dominated by continual escape from accumulating population immunity, driving ongoing antigenic change, where some viruses are better able than others to sustain adaptive evolution [1]. Adaptation of either kind leaves a signature in amino-acid replacing nonsynonymous vs silent synonymous substitutions, with methods ranging from simple comparisons of nonsynonymous to synonymous substitution rates (dN/dS) to McDonald–Kreitman-

style approaches [2] that weigh mutations fixed along a virus's successful trunk lineage against those lost on unsuccessful side branches [3]. Such approaches have revealed rapid, continued adaptation in the SARS-CoV-2 spike S1 subunit [4, 5].

A complementary class of methods estimates fitness directly from the dynamics of variant frequencies rather than from the composition of mutations [6]. Multinomial logistic regression (MLR) models the frequencies of co-circulating variants through time and infers a relative growth rate, or fitness, for each [7, 8]. Because it expresses fitness as a difference in growth rate between variants, this measure maps directly onto the population-genetic notion of selective advantage. These growth-rate differences correspond to differences in the time-varying effective reproduction number between co-circulating variants [9]. Aggregating these per-variant fitnesses into the rate of change of mean population fitness yields the population's fitness flux [10], a direct, frequency-based alternative to dN/dS for quantifying the tempo of adaptation.

Here we use this frequency-based view of fitness to trace how SARS-CoV-2 has adapted from the early pandemic in 2020 through 2025, spanning the transition from initial host adaptation to sustained evolution for antigenic novelty. We place the rate of SARS-CoV-2 fitness change in context by comparing it against seasonal influenza A/H3N2 which exhibits canonically rapid and continuous adaptation [11]. Finally, we relate the inferred changes in fitness to molecular predictors, most directly the accumulation of spike mutations, to identify the substitutions that drive fitness gain.

Results and discussion

Frequency dynamics

We follow population genetics first principles to compute the frequency through time of a haploid allele under deterministic selection in a fixed-generation Wright–Fisher population. If an allele is at frequency x, then after a single generation with selective advantage s the expected allele frequency will be

$$x' = \frac{x(1+s)}{x(1+s)+(1-x)}$$

Compounded over t generations, the expectation from initial frequency p follows

$$x(t) = \frac{p(1+s)^t}{p(1+s)^t+(1-p)}$$

Generalizing this two-allele model to n co-circulating variants, each with initial frequency p_i and selective advantage s_i, variant i's frequency is its relative abundance normalized by the sum across all variants,

$$x_i(t) = \frac{p_i(1+s_i)^t}{\sum_j p_j(1+s_j)^t}$$

Moving from many discrete generations to continuous time, $(1+s_i)^t \approx \exp(t\log(1+s_i))$, so writing the growth rate $f_i = \log(1+s_i)$ gives the probability that a virus sampled at time t is labeled as variant i

$$\Pr(X=i) = x_i(t) = \frac{p_i \exp(f_i t)}{\sum_j p_j \exp(f_j t)}$$

This is the multinomial logistic regression (MLR) model, which has been widely used for modeling SARS-CoV-2 variant frequencies [7, 8]. The denominator normalizes the exponential growth/decay of individual variants so that overall frequency sums to 1. The model has $2n$ parameters, with each variant i having an initial frequency p_i and a fixed growth rate f_i.

Because growth rates are necessarily relative, we fix an arbitrary "pivot" variant as a reference with growth rate $f = 0$. MLR growth rates are directly estimated in terms of calendar time with per-day or per-year values of f. To express fitness in pergeneration units we multiply each per-day rate by the generation time τ measured in days, giving $f_i = \tau f_i^{\text{day}}$, the change in log frequency accrued over a single generation. We assume τ of 5.0 days for pre-Omicron SARS-CoV-2, 3.2 days for post-Omicron SARS-CoV-2 and 3.2 for influenza H3N2 (see Methods). Throughout, we refer to this per-generation growth rate $f_i = \log(1 + s_i)$ as the fitness of variant i, or equivalently $\exp(f_i) = 1 + s_i$ is its pergeneration multiplicative fitness, the factor by which the variant's abundance grows each generation relative to the pivot. Because f_i is defined on a log scale, mean fitness, fitness variance, fitness flux and changes in fitness between lineages are all likewise computed on this log scale.

We estimate frequencies and fitnesses of SARS-CoV-2 clades in 1-year sliding windows between Jan 2020 and Dec 2025 (Figure 1). In each window we collect clade sequence counts for viruses sampled from the USA and estimate per-variant frequencies and fitnesses. We use only the USA, the one country with sufficient temporal sequencing coverage over this period. We collapse rare clades into a single "other" clade for MLR analysis to prevent noisy estimates from low sequence counts (see Methods). The match between the empirical frequencies (dotted trajectories) and MLR frequencies (solid trajectories) indicates the model fits well despite having few parameters.

For comparison purposes, we take a similar approach to estimating frequencies and fitnesses of seasonal influenza H3N2 (Figure 2). Here we use 2-year sliding windows to account for slower frequency dynamics in seasonal influenza and still only use data from the USA. The model fits are worse for H3N2 compared to SARS-CoV-2. This is especially apparent at junctions between influenza seasons where stochastic seeding of a new season may result in a discontinuity of clade frequency compared to MLR expectation. However, H3N2 fits remain sufficient to estimate the magnitude of fitness effects.

We "scaffold" MLR fitness estimates across windows to arrive at a single fitness estimate per variant. Here, each window only measures fitness differences, with its own arbitrary zero. We solve for the single set of clade fitnesses and per-window offsets that best fit every window at once, weighting by abundance (see Methods).

Fitness flux

With variant frequency $x_i(t)$ and constant variant fitness f_i, we describe the mean population fitness as a standard weighted sum $f(t) = \sum_i x_i(t) f_i$. The fitness flux [10] of the population is then the rate of change of population fitness at a given time $\phi(t) = \Delta f(t)/\Delta t$.

Integrating this rate gives the cumulative fitness flux

$$\Phi(t) = \int_{t_0}^{t} \phi\left(t'\right) dt' = \bar{f}(t) - \bar{f}(t_0)$$

the total adaptive change accumulated along the population's trajectory. Because variant fitnesses are estimated only relative to a pivot, an individual variant's scaffolded fitness is meaningful as a difference from a baseline rather than as an absolute value. Chaining these locally-measured advantages across overlapping windows places variant i at a cumulative fitness flux $\Phi_i = f_i - f_0$ relative to the founding variant, and the population sits at the frequency-weighted average $\Phi(t) = \sum_i x_i(t)\, \Phi_i$.

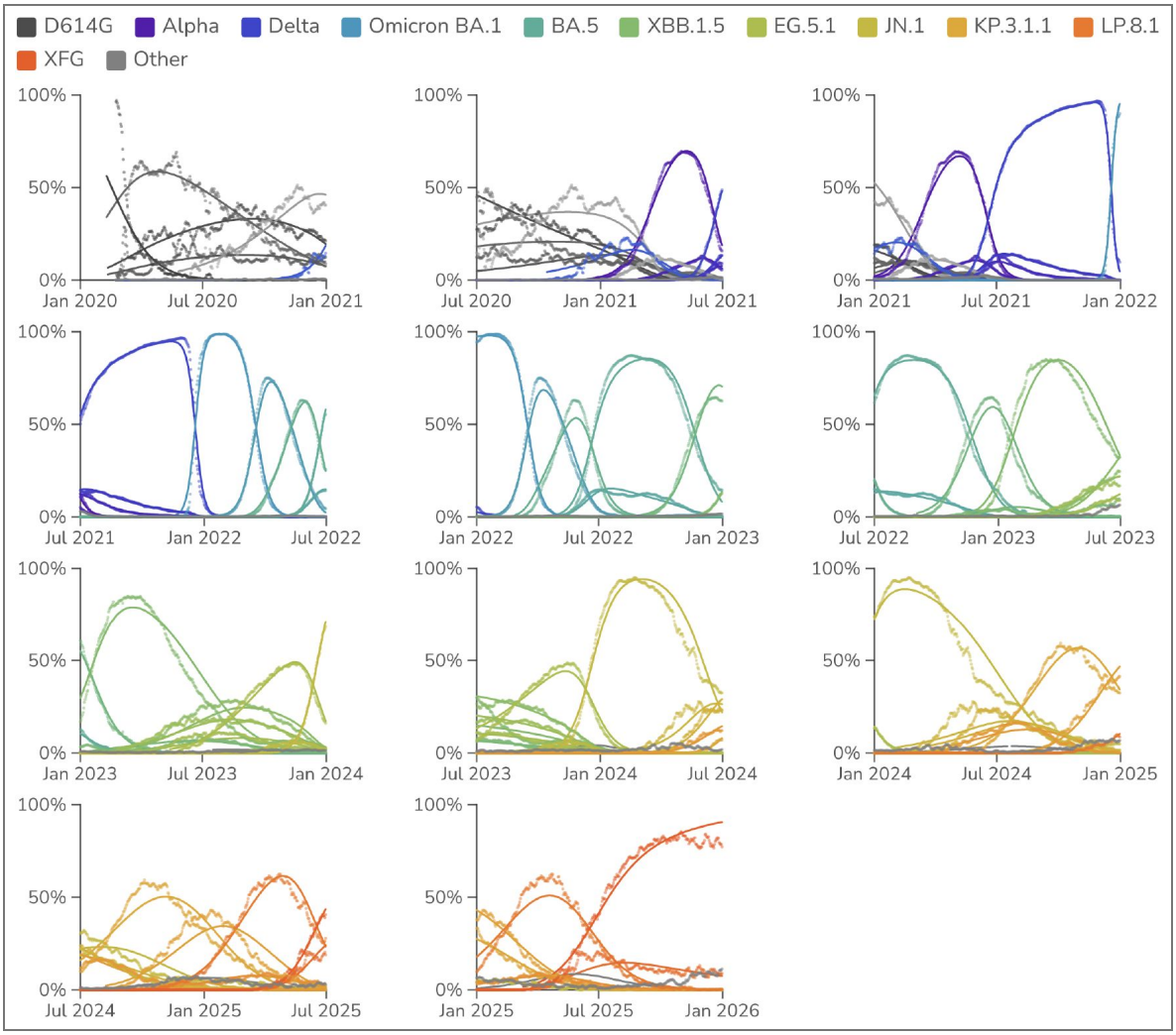


Figure 1. **Relative frequencies of SARS-CoV-2 clades through time.**

Points represent empirical frequencies of SARS-CoV-2 Nextstrain clades, while solid lines represent modeled frequencies from multinomial logistic regression (MLR). All data is taken from the USA. The MLR analysis assumes that the fitness of each clade is constant through time. MLR frequency lines are drawn where there is sufficient sequence data to estimate empirical frequencies.

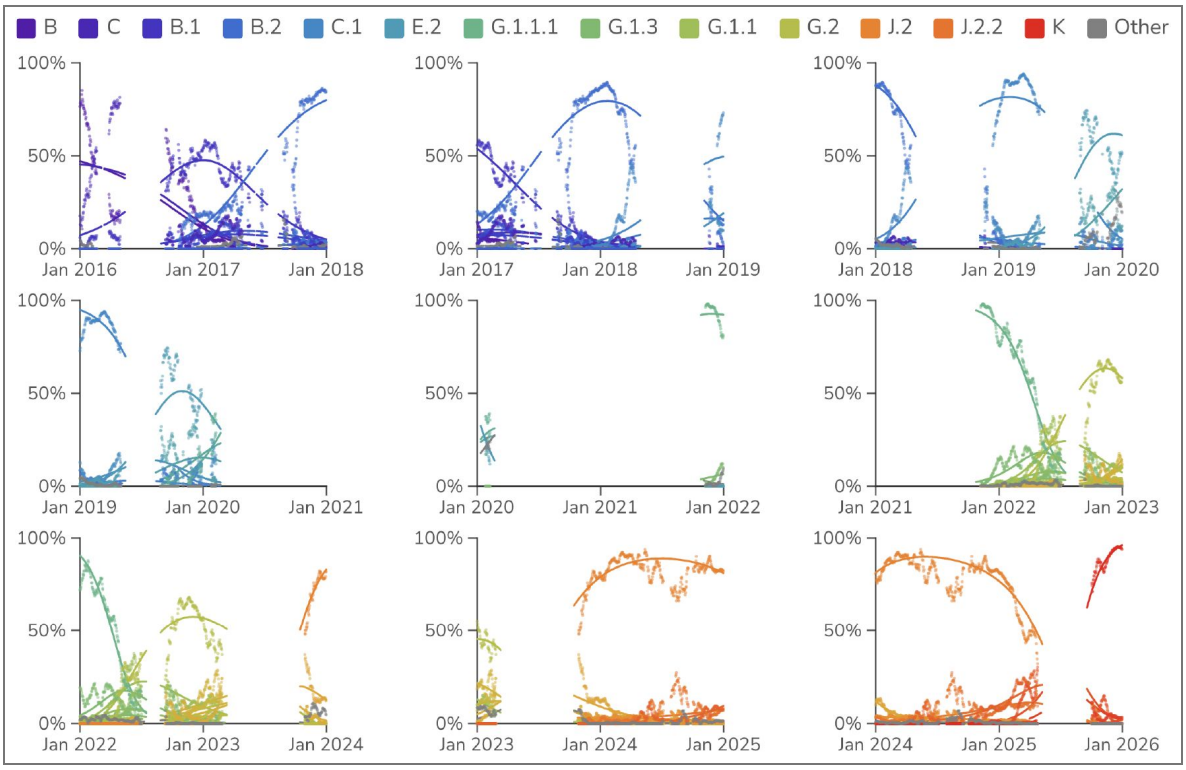


Figure 2. **Relative frequencies of H3N2 clades through time.**

Points represent empirical frequencies of H3N2 Nextstrain clades, while solid lines represent modeled frequencies from multinomial logistic regression (MLR). All data is taken from the USA. The MLR analysis assumes that the fitness of each clade is constant through time. MLR frequency lines are drawn where there is sufficient sequence data to estimate empirical frequencies.

We find that SARS-CoV-2 initially accumulated fitness flux rapidly with mean fitness doubling every 6 months between Jan 2021 and Jun 2022, but then slowing to doubling every 2.4 years from July 2022 to Dec 2025 (Figure 3). After initial spread of D614G [12] in 2020, we observe a lull, followed by rapid growth in fitness in 2021 and 2022 with initial VOCs, Omicron and initial Omicron sub-lineages [13], and then a slower, more steady pace since 2024. There is a mix of large jumps in fitness (familiar examples like Delta and BA.1, but also more recently with JN.1) and smaller, more gradual step change.

Seasonal influenza H3N2 shows a fundamentally similar pattern of emergence of new clades and their replacement of existing diversity. However, H3N2 dynamics play out on a slower timescale (Figure 4). Rather than SARS-CoV-2's months-scale doubling, H3N2 adapts much more slowly, doubling roughly every 10.0 years over the course of 2016 to 2025. Greater coexistence of multiple co-circulating clades is also apparent relative to SARSCoV-2.

SARS-CoV-2 clade frequencies and fitnesses can be viewed as a phase portrait, plotting each clade's empirical frequency against its fitness relative to the daily population average (Figure 5). A clade emerges at low frequency and high relative fitness, sweeps up in frequency as its relative fitness declines toward the population average, peaks near a relative fitness of zero, and then falls back to low frequency as it is outcompeted. Clades that start out with a greater advantage over the population average tend to sweep to higher maximum frequency than clades that start with less of an advantage.

Fisher's fundamental theorem

Multistrain models that allow for antigenic evolution produce traveling waves in antigenic space [14]. More broadly, many mutations of small fitness effect create traveling fitness waves where the rate of advance toward higher fitness is proportional to the variance in fitness [15]. This is a consequence of Fisher's fundamental theorem of natural selection

$$\frac{df}{dt} = \mathrm{Var}(f)$$

where "the rate of increase in fitness of any organism at any time is equal to its genetic variance in fitness at that time" [16].

We can investigate this relationship directly in SARS-CoV-2 (Figure 6), where we find that timepoints with larger variance in fitness $\mathrm{Var} f(t)] = \sum_i x_i(t)(f_i - f(t))^2$ correlate well with timepoints with larger change in mean population fitness $\Delta f(t)/\Delta t$. In fact we find that the relationship is near the 1:1 expectation from Fisher's theorem (slope = 1.10, Pearson r = 0.94). Looking in detail at rate of fitness flux through time, we find a yearly average fitness flux of 18.0 $\times 10^{-3}$ per-gen in 2021 followed by a reduction to 1.7 – 1.8 $\times 10^{-3}$ per-gen in 2024 and 2025. This shows that the rate of adaptation of SARS-CoV-2 has been slowing as low hanging fruit of host adaptation is exhausted, leaving only red-queen antigenic evolution to drive adaptation.

Compared to SARS-CoV-2, influenza H3N2 shows generally lower rates of fitness flux, averaging 0.5 $\times 10^{-3}$ per-gen from 2016 to 2025 (Figure 7). This is roughly 3 times lower than recent years of SARS-CoV-2 fitness flux. However, it remains possible that SARSCoV-2 slows further in the coming years.

For both SARS-CoV-2 and influenza H3N2, the connection between fitness flux and fitness variance is clear. This suggests from first principles that interventions that decrease variance in fitness across the virus population would be expected to slow adaptation, while interventions that increase variance would be expected to speed adaptation.

Mutational fitness effects

A simple correlation of cumulative mutations against cumulative fitness flux will fail due to phylogenetic non-independence, and instead we rely on phylogenetic contrasts of parent and child lineages [17]. Pango lineages [18] provide a convenient granular and hierarchical nomenclature well suited to this.

Figure 3. **Cumulative SARS-CoV-2 fitness flux.**

Empirical frequencies of SARS-CoV-2 clades are represented by vertical thickness and placement on the y-axis represents cumulative fitness flux estimated from multinomial logistic regression (MLR). Gray lines are least-squares fits to the mean fitness over each labeled period, annotated with the implied doubling time. All data is taken from the USA. The MLR analysis assumes that the fitness of each clade is constant through time.

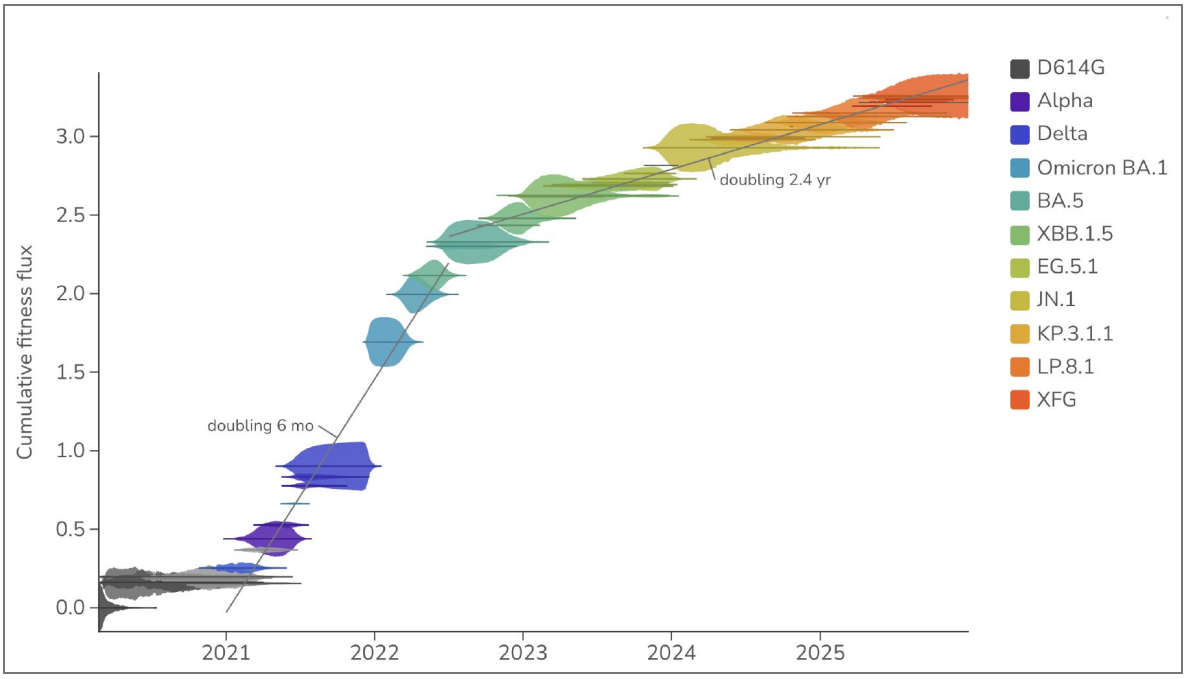


Figure 4. **Cumulative H3N2 fitness flux.**

Empirical frequencies of H3N2 clades are represented by vertical thickness and placement on the y-axis represents cumulative fitness flux estimated from multinomial logistic regression (MLR). Gray lines are least-squares fits to the mean fitness over each labeled period, annotated with the implied doubling time. All data is taken from the USA. The MLR analysis assumes that the fitness of each clade is constant through time.

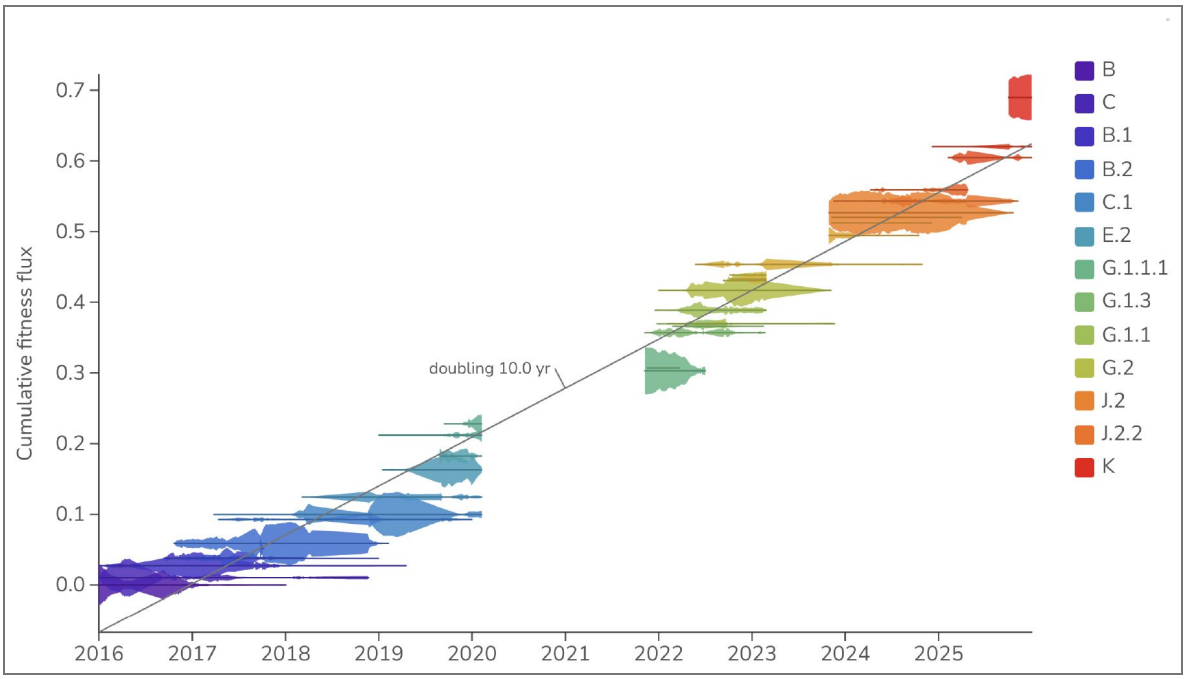


Figure 5. **Frequency vs fitness phase portrait for SARS-CoV-2 clades.**

Each line traces a SARS-CoV-2 clade's trajectory over time through empirical frequency (x-axis, logit scale) and fitness relative to the daily population average (y-axis), estimated from multinomial logistic regression (MLR). All data is taken from the USA. The MLR analysis assumes that the fitness of each clade is constant through time.

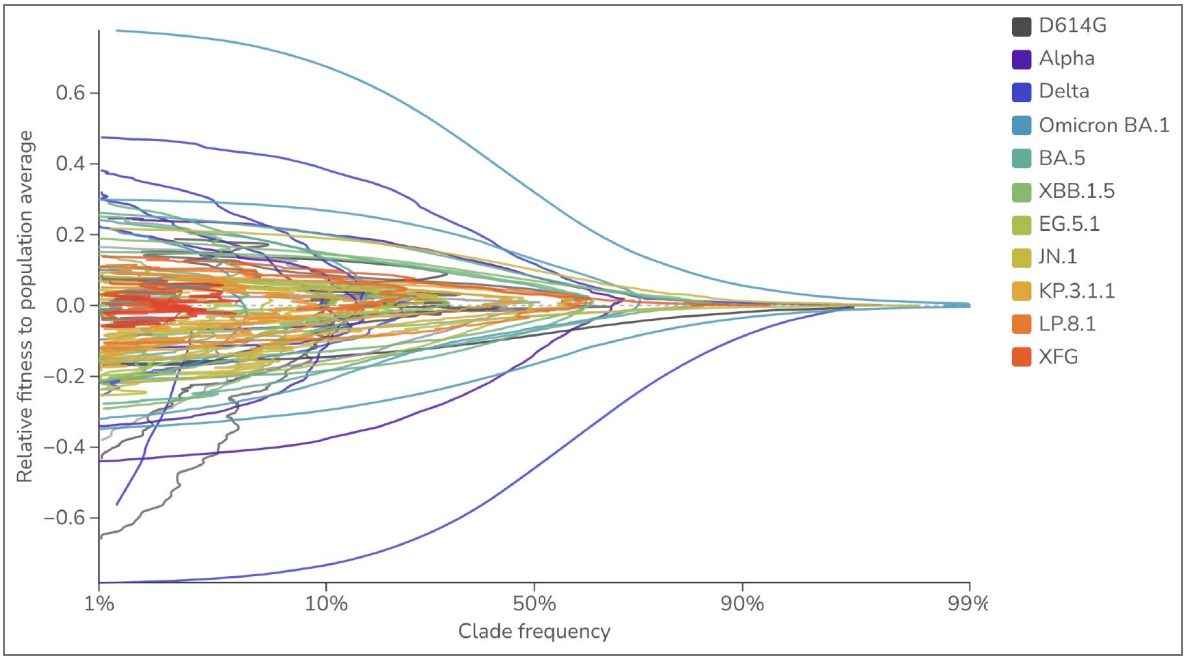
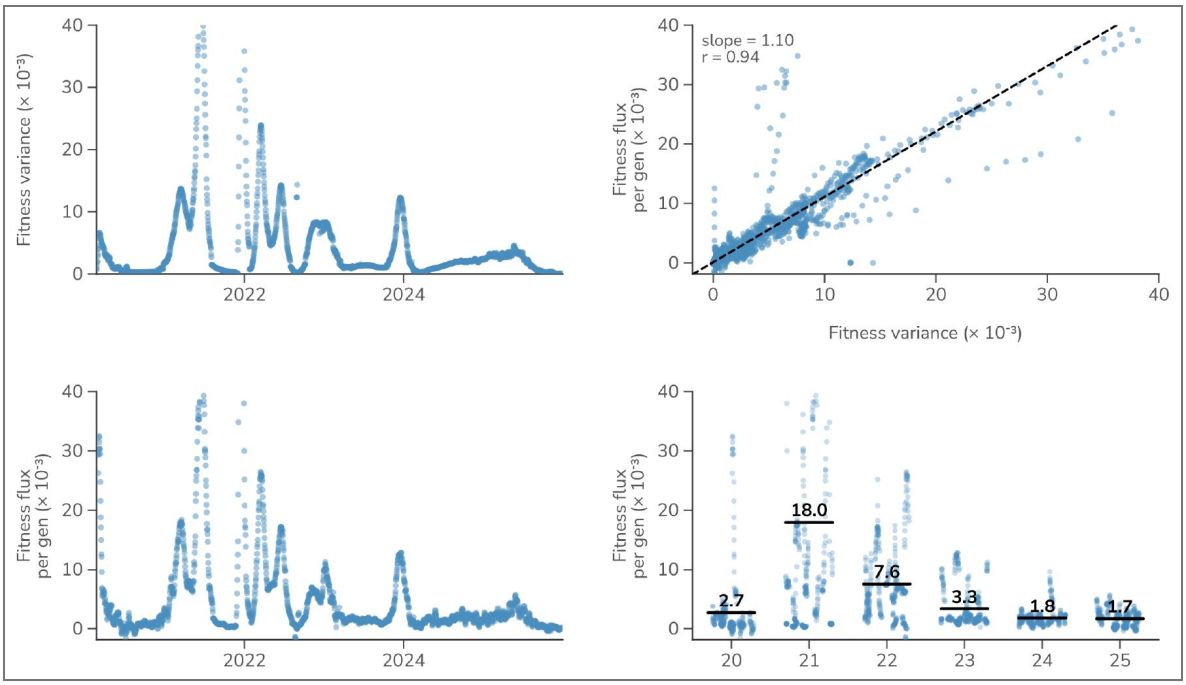


Figure 6. **Fitness variance and fitness flux in SARS-CoV-2.**

Fitness variance is compared to fitness flux, where each dot represents a daily timepoint.



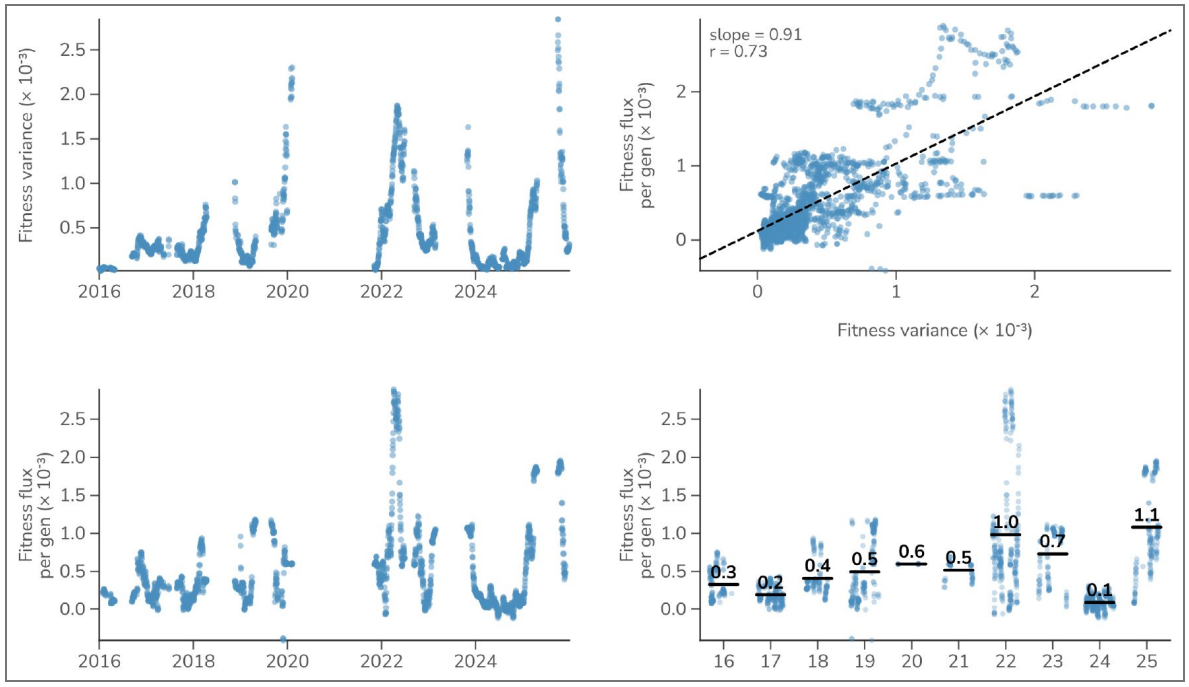


Figure 7. Fitness variance and fitness flux in H3N2.

Fitness variance is compared to fitness flux, where each dot represents a daily timepoint.

We define Pango parent-to-child branches by finding mutations that accumulate between hierarchical Pango lineages. Given estimates of per-lineage fitness, we also calculate the change in fitness between each parent and child. As expected, we observe that Pango lineages are granular with parent-to-child changes in spike mutations, non-spike mutations and fitness being modest, with most branches adding only a handful of substitutions and shifting fitness by a small amount (Figure 8).

We compare the change in fitness along each parent-to-child branch against the substitutions it acquired in different regions of the SARS-CoV-2 genome (Figure 9). Spike S1 substitutions and those in the receptor-binding domain (RBD) in particular, carry the strongest positive association with fitness gain, while ORF1ab and accessory gene substitutions show lower regression slopes and correlation coefficients. Splitting branches into early and late periods shows that the fitness value of a given substitution is largest early and decays over time.

We can track this relationship through time by fitting the regression separately within each season (Figure 10). The per-substitution effect on fitness of spike and RBD changes is largest early and erodes toward zero as the readily accessible routes to host adaptation are exhausted, while other regions stay near zero throughout. However, across accessory proteins and in ORF1ab there is moderate marginal correlation from 2020 to 2022 between substitutions and fitness change; the multiple regression below shows this to be a confound of co-occurrence with spike rather than an independent effect. Although per-substitution effect (ie regression slope) of RBD decays from 2020, the predictive ability of spike S1 and RBD substitutions as measured by Pearson and Spearman correlations stays high through the period with average correlation coefficients of $r = 0.72$ and $\rho = 0.54$ for spike S1 and $r = 0.70$ and $\rho = 0.53$ for spike RBD.

The marginal relationships in Figure 9 cannot on their own establish which substitutions drive fitness. A more evolved lineage accumulates more substitutions across the whole genome, so a region can correlate with fitness merely by co-varying with a genuinely causal region. The moderate marginal association of ORF1ab substitutions is a case in point.

To isolate each region's independent contribution we fit a multiple linear regression of the change in substitution count across four non-overlapping genome regions to per-branch change in fitness (Figure 11). Once spike is controlled for, most of the positive signal sits in the RBD ($\beta = 0.064$ per substitution, $p < 10^{-20}$), with a smaller contribution from the remainder of S1 ($\beta = 0.015$, $p = 1.5 \times 10^{-4}$), while the ORF1ab and accessory-protein coefficients collapse to near zero and are no longer distinguishable from no effect ($\beta = 0.002$, $p = 0.32$ and $\beta = 0.003$, $p = 0.41$, respectively). The apparent marginal association of ORF1ab is therefore a confound of its co-occurrence with spike change rather than evidence that ORF1ab substitutions themselves raise fitness. The four-region model explains most of the variance in per-pair fitness change ($R^2 = 0.73$), with predicted and observed changes falling along the 1:1 line.

Predicting fitness effects

The core idea of comparing change in mutation count to change in fitness expresses a similar logic to McDonald-Kreitman tests [2] where the key comparison is relative success of lineages bearing different mutation patterns. This predictor-vs-growth-rate formulation [4] should be robust to many confounders that affect other measures of adaptation.

We compare our simple, non-parameterized predictor of relative spike S1 substitution count against two recent deep-learning proposals for scoring evolutionarily successful substitutions (Figure 12). These are EvEscape [19] and protein-language-model semanticity [20], the latter reimplemented with pretrained and fine-tuned ESM-2 embeddings [21] (see Methods). ESM is fine-tuned to SARS-CoV-2 spike sequences from 2020 through 2022. Both perform similarly to a plain count of spike S1 substitutions in disambiguating the fitness of co-circulating SARS-CoV-2 lineages.

Conclusions

Most measures of viral adaptation are indirect, diagnosing the presence of selection based on mutation patterns. Here we instead read adaptation directly off the dynamics of variant frequencies, aggregating per-variant growth rates into the population's fitness flux. This turns the

Figure 8. **Distributions of mutation and fitness change across SARS-CoV-2 lineage branches.**

Across all parent-to-child Pango lineage branches, the change in the number of spike substitutions, the change in non-spike substitutions, and the change in fitness. Each bar is the fraction of branches in that bin; rare large founder jumps fall beyond the plotted range.

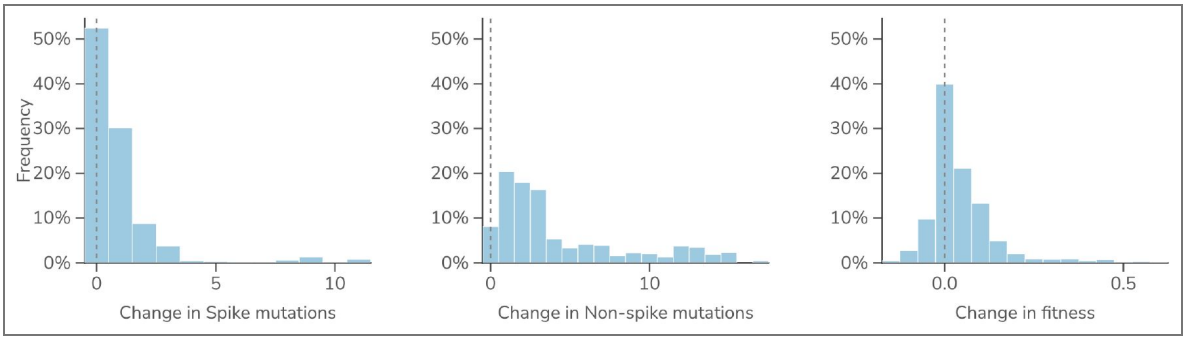


Figure 9. **Lineage-specific amino acid change versus lineage-specific fitness change across regions of the SARS-CoV-2 genome.**

Each point is one parent-to-child Pango lineage branch in one season: the change in the number of substitutions in a genome region (x) against the change in fitness (y), colored by time from blue (2020) to red (2025), with a least-squares fit per panel.

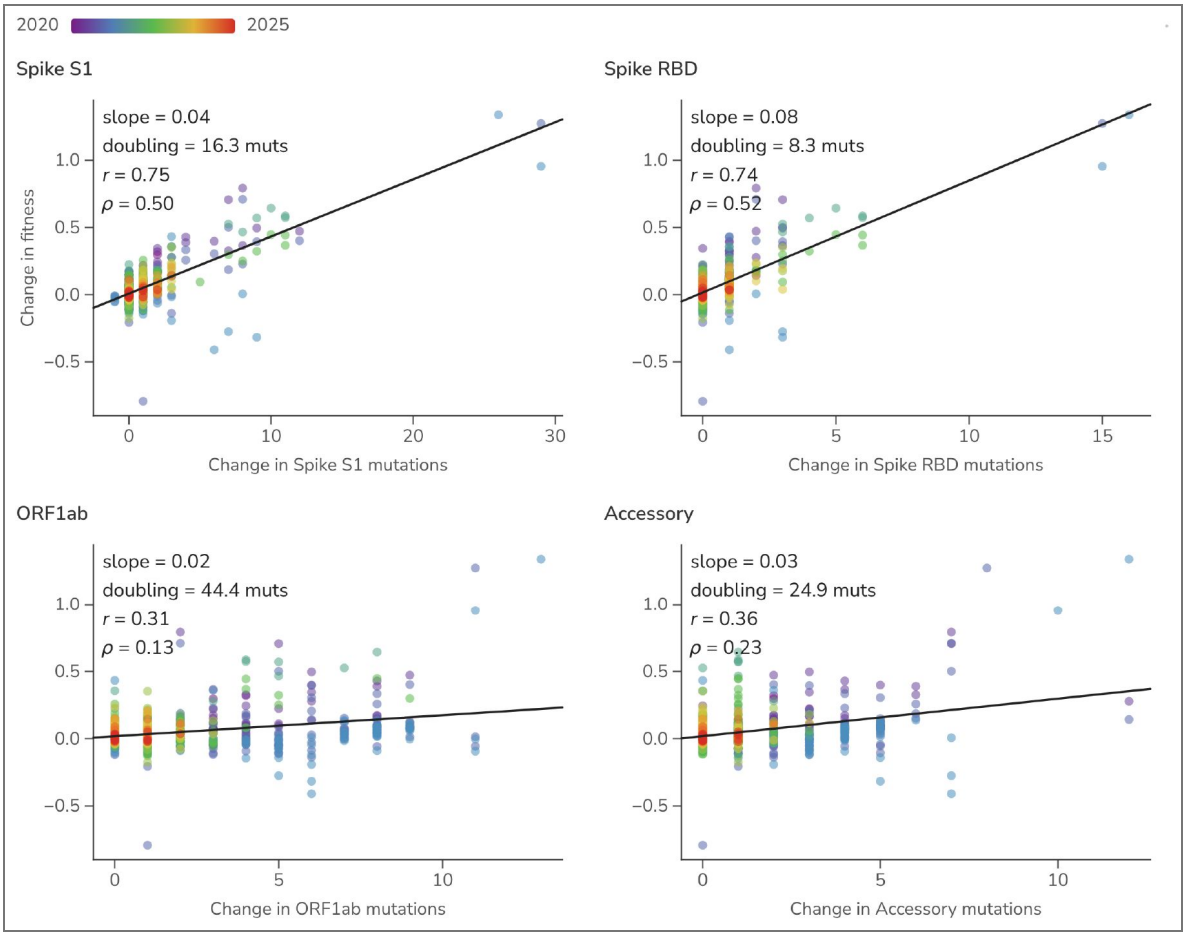


Figure 10. Strength of the mutation–fitness relationship through time.

For each season the parent-to-child branches are summarized by regression slope relating change in regional sub-stitutions to change in fitness, with one line per genome region.

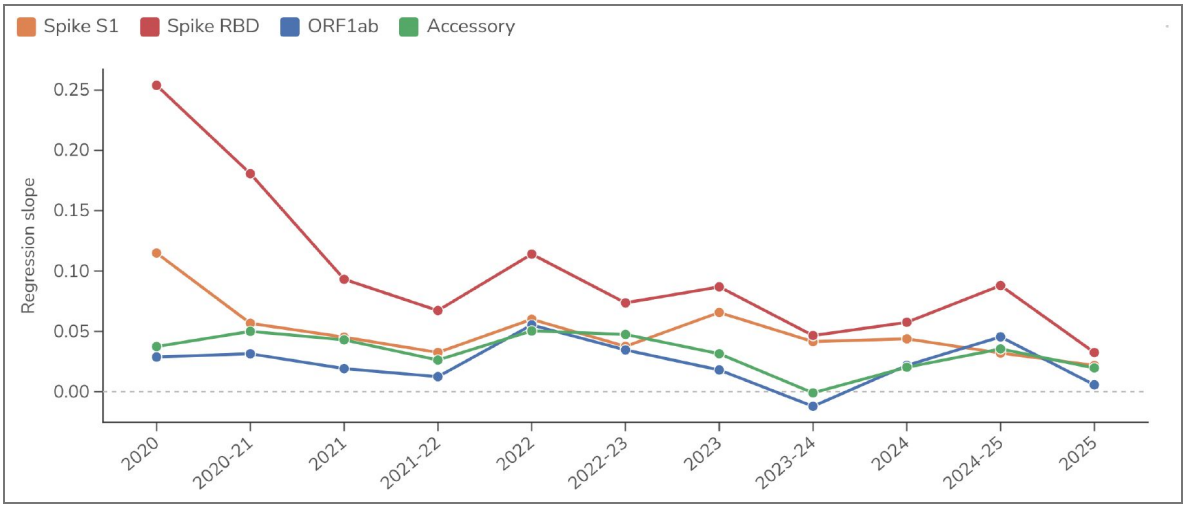
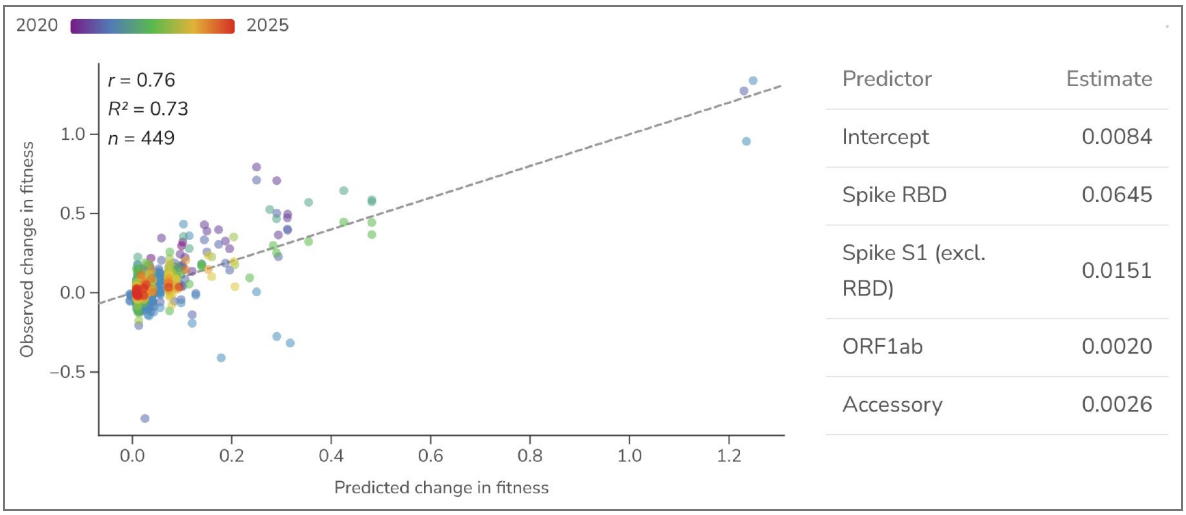


Figure 11. Multiple regression of fitness change on non-overlapping genome regions.

An ordinary-least-squares fit of the change in fitness on the change in substitution count in four non-overlapping regions of the SARS-CoV-2 genome (spike RBD, spike S1 outside the RBD, ORF1ab, accessory), fit over unique parent-to-child pairs (collapsing observations of a pair that recur across windows). The table gives each region's partial estimate. The scatter plots each parent-to-child Pango lineage branch's model-predicted change in fitness (x) against its observed change in fitness (y), colored by time from blue (2020) to red (2025) and with dashed 1:1 calibration line.



Predictor	Estimate
Intercept	0.0084
Spike RBD	0.0645
Spike S1 (excl. RBD)	0.0151
ORF1ab	0.0020
Accessory	0.0026

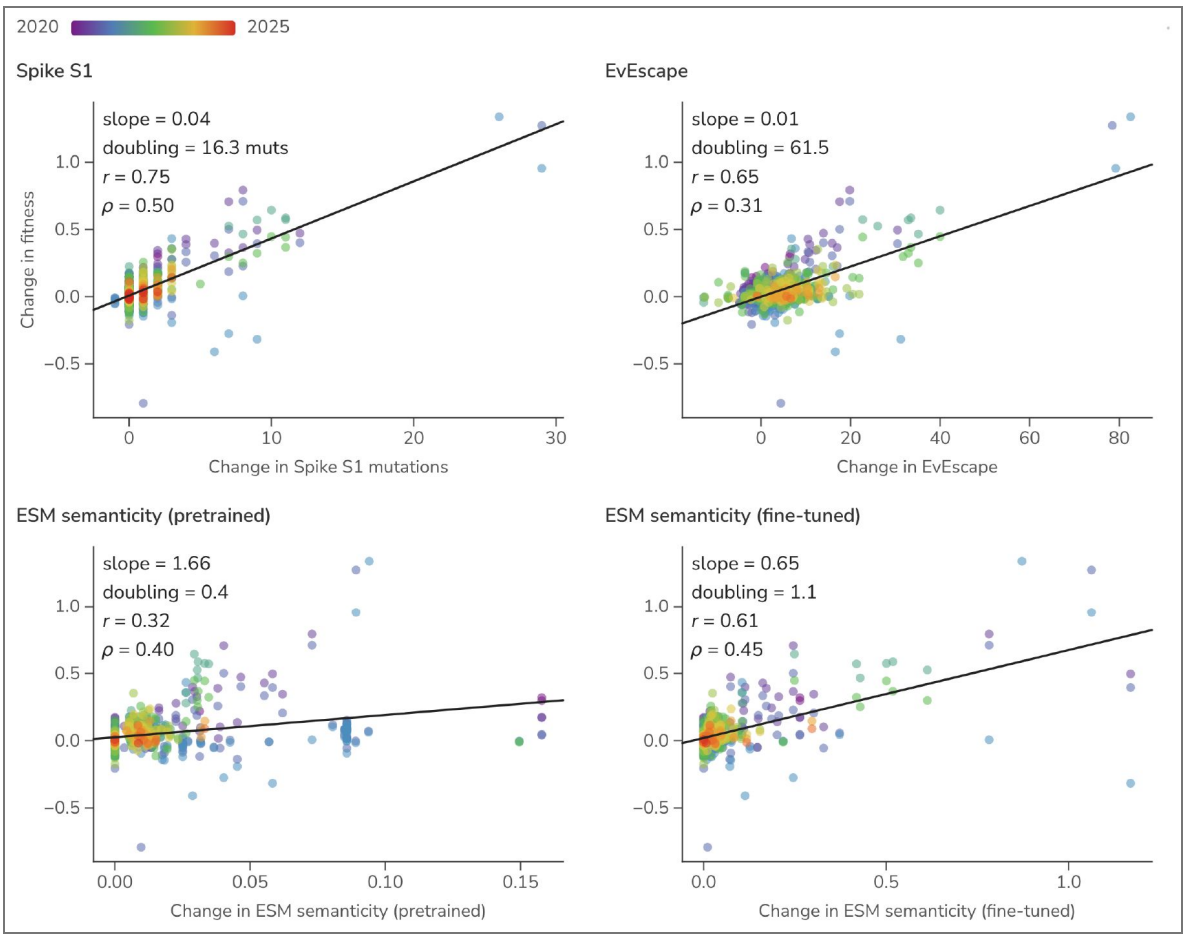


Figure 12. Lineage-specific predictors versus lineage-specific fitness change.

Each point is one parent-to-child Pango lineage branch in one season: change in predictor value (x) against the change in fitness (y), colored by time from blue (2020) to red (2025), with a least-squares fit per panel.

tempo of adaptation into a single quantity that can be followed through time, placed on a common per-generation scale across pathogens and connected to first principles. On this scale SARS-CoV-2 adapts rapidly, doubling in fitness roughly every 6 months during initial variant emergence in 2021 and early 2022, but decelerating towards a baseline flux in 2024, while seasonal H3N2 sustains a slower, steadier flux. Importantly, these numbers carry an absolute scale and connect back to epidemiological impact [9].

The generality of the approach rests on a single requirement: a way to bin genetic diversity into discrete, comparable variants. For SARS-CoV-2 and influenza this comes off the shelf, with Nextstrain clades supporting the frequency and flux analysis and finer, hierarchically nested Pango lineages supporting the phylogenetic-contrast analysis of mutational effects. Pathogens without an established nomenclature could be analyzed via automated methods that partition a tree into lineages [22, 23].

Beyond describing historical adaptation, the per-branch contrast of mutation against fitness change yields a simple, interpretable account of which substitutions matter. For SARSCoV-2 the signal concentrates in spike (and particularly spike RBD) and a plain count of spike S1 substitutions disambiguates the relative fitness of co-circulating lineages about as well as recently proposed deep-learning escape and protein-language-model scores [19, 20]. This makes the mutation-to-fitness deltas a strong and transparent baseline for forecasting variant success, where a predictor that does not improve on counting spike substitutions has not yet justified its added complexity.

Methods

Sequence data

For SARS-CoV-2, we use curated “open” data from Nextstrain [24] that draws from NCBI GenBank. For influenza H3N2, we use data from GISAID [25]. In each case, the raw sequences are processed with Nextclade [26] to assign Nextstrain clade and Pango lineage to [18] SARS-CoV-2 sequences and to assign subclade [27] to influenza H3N2 sequences. We filter out sequences with Nextclade overall QC status of “bad”. We additionally filter to sequences collected from the USA. This leaves 3,588,802 total sequences for SARS-CoV-2 sampled between 2020 and 2025 and 44,456 total sequences for H3N2 sampled between 2016 and 2025.

Multinomial logistic regression

We conducted multinomial logistic regression (MLR) using the evofr package (*github.com/blab/evofr*) on 1-year sliding windows for SARS-CoV-2 (11 windows total) and 2-year sliding windows for H3N2 (9 windows total). For each window we treat each clade as a distinct variant, collapsing rare clades together into a single “other” category before fitting. For both SARSCoV-2 and H3N2, a clade is modeled separately only if it reaches at least 50 sequences and a mean frequency of at least 0.1% within the window, while clades below either threshold are merged into “other”. This leaves between 7 and 18 clades per window (median 15) for SARS-CoV-2 and between 5 and 13 (median 9) for influenza H3N2.

We use generation time τ of 5.0 days for pre-Omicron SARS-CoV-2 following [28–30], generation time of 3.2 days for post-Omicron SARS-CoV-2 following [31,32] and generation time of 3.2 days for seasonal influenza H3N2 following [33–35]. At first order, the fitness $f_i = \tau f_i^{\text{day}}$ is the log ratio of reproduction numbers between variant i and the pivot, $\log(R_i/R_{\text{pivot}})$. Only the mean generation interval τ enters at this order. The shape of the generation-interval distribution enters at second order, informed by the interval’s variance and absolute per-day growth rates rather than f_i^{day}. Because the frequency data identify only this relative rate, the second order correction cannot be estimated from frequency dynamics alone. The per-generation multiplicative fitness $\exp(f_i) = 1 + s_i$ therefore equals the reproduction-number ratio R_i/R_{pivot} under a fixed (point) generation interval and is an upper bound under a realistically dispersed interval [36], for which the fuller frequency-to-reproduction-number conversion is given in Figgins and Bedford [9].

We conduct a parallel MLR analysis of SARS-CoV-2 Pango lineages. Because lineages are hierarchically nested, rather than collapsing rare lineages into a shared "other" we roll each lineage with fewer than 500 sequences up into its parent lineage, repeating until every retained lineage clears this count. A lineage is additionally retained only if at least 200 sequences are assigned to that lineage itself rather than to a descendant sub-lineage, otherwise it is folded into "other". This leaves between 20 and 165 lineages per window (median 79) for SARS-CoV-2. Rationale for specific collapse cutoffs is available at *github.com/blab/fitness-flux/tree/main/inclusion-thresholds*.

Scaffolding across timepoints

Within each sliding window the MLR model estimates each variant's fitness only relative to that window's pivot, so every window carries its own arbitrary additive zero and the perwindow estimates $f_{i,w}$ are not directly comparable. We recover a single fitness per variant by treating scaffolding as a weighted two-way additive model: each estimate is a variant effect minus a window effect, $f_{i,w} \approx f_i - c_w$, where f_i is variant i's global fitness and c_w is window w's offset. We choose the f_i and c_w that jointly minimize the abundance-weighted squared error across every window,

$$\min_{\{f_i\},\{c_w\}} \sum_{i,w} a_{i,w}(f_{i,w} - f_i + c_w)^2$$

weighting each estimate by $a_{i,w}$, the area under variant i's modeled-frequency curve in window w, so a window in which a variant is rare with poorly constrained MLR estimate will contribute negligibly. The optimum is a pair of interleaved abundance-weighted means, each variant's fitness being the weighted mean of its offset-corrected estimates over its windows

$$f_i = \frac{\sum_w a_{i,w}\,(f_{i,w} + c_w)}{\sum_w a_{i,w}}$$

and each window's offset the weighted-mean gap between the global scale and that window's estimates

$$c_w = \frac{\sum_i a_{i,w}\,(f_i - f_{i,w})}{\sum_i a_{i,w}}$$

We solve by alternating the two to convergence. The overlap of variants between windows ties them into one connected scale, leaving a single global constant free, which we fix by shifting all values so that the founding variant, our least-fit baseline, sits at zero, leaving each variant's scaffolded value as its cumulative fitness flux $\Phi_i = f_i - f_0$.

Lineage mutation counts and branch contrasts

To relate change in fitness to change in genotype, we count amino-acid substitutions per Pango lineage and compare each lineage against its parent. Per-lineage substitution counts are read from the Nextclade SARS-CoV-2 reference tree at *nextstrain.org/nextclade/nextstrain/sars-cov-2/wuhan-hu-1/orfs*, in which each tip corresponds to a Pango lineage. For each lineage we count substitutions relative to the Wuhan-Hu-1 reference. We tally these by region: the spike S1 subunit, the receptor-binding domain (RBD, 319–541) within S1, ORF1ab, and the accessory and structural genes (ORF3a, E, M, ORF6, ORF7a, ORF7b, ORF8, N).

We then form parent-to-child branches between hierarchically nested Pango lineages. Within each window a lineage's parent is its closest retained ancestor, where the retained set is fixed by the collapsing described above, so lineages that were rolled up into a parent or folded into "other" do not themselves appear as branch endpoints. For every branch whose parent and child both carry an MLR fitness estimate in that window, we record the change in substitution count in each

genome region and the change in fitness, taken as the difference in their per-window MLR fitness; because both endpoints are estimated against the same window's pivot, this contrast is well defined without scaffolding. A branch is recorded once per window in which it appears, so a lineage pair that co-circulates across several windows contributes several observations. These per-branch deltas in mutation count and fitness are the unit of the mutational-fitness analyses. For the multiple regression we collapse these repeated observations to a single contrast per parent-to-child pair, averaging the per-window fitness change, so that coefficients and *p*-values are not inflated by this pseudoreplication.

Mutational fitness predictors

We compare the per-branch substitution counts above against two externally proposed predictors of mutational fitness, each reduced to a per-lineage value and contrasted across the same parent-to-child branches. EvEscape [19] combines a variational-autoencoder fitness model with residue accessibility and biochemical dissimilarity to score spike mutations. We use the precomputed all-strain EvEscape scores released at *evescape.org/data* and take each Pango lineage's score as the mean EvEscape across the sequences assigned to that lineage.

Semanticity follows the semantic-change measure of Hie et al. [20], reimplemented with ESM-2 [21]. Each Pango lineage's spike amino-acid sequence is embedded with the 650M-parameter ESM-2 model (`esm2_t33_650M_UR50D`), taking the CLS-token representation of the final (33rd) layer as a 1280-dimensional sequence embedding. We embed lineages with both the released pretrained weights and weights fine-tuned under a masked-language-model objective (15% of residues masked) for one epoch (AdamW, learning rate 5×10^{-5}) on roughly 16,000 SARS-CoV-2 spike sequences collected from 2020 through 2022. Finetuning and embedding code is available at *github.com/blab/embedded-pathways*. The semanticity of a branch is the Euclidean distance between its child and parent lineage embeddings.

Reproducibility

The workflow to reproduce data, analyses and manuscript is available at *github.com/blab/fitness-flux*. SARS-CoV-2 sequence data were sourced from publicly available open data in NCBI GenBank and so this side of the workflow is easily runnable. However, influenza H3N2 sequence data was collected from GISAID and these data are not allowed to be reshared and consequently the influenza side of the workflow is unfortunately not reproducible without separately providing GISAID data.

Data availability

This work uses publicly available viral genome sequences. SARS-CoV-2 genome sequences are sourced from publicly available at GenBank and is trivially reproducible via GitHub workflow at *https://github.com/blab/fitness-flux*. Seasonal influenza H3N2 are sourced from *https://gisaid.org/* and users are not permitted to reshare data. Influenza workflow is provided in GitHub, but the source data will need to be sourced directly to reproduce findings.

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Peer reviews

Reviewer #1 (Public review):

Summary:

This manuscript calculates a measure of fitness flux for influenza and SARS-CoV-2 viruses, normalising by a generation time go generate comparable measures of viral adaptation over time. Potential correlates with gain in fitness flux, such as the number of mutations in different regions of the virus, are evaluated in terms of their correlation with the measured gain. A count of mutations in the S1 subunit of Spike performs slightly better in explaining observed fitness changes than previous deep-learning models describing SARS-CoV-2 evolution.

Strengths:

I enjoyed reading this manuscript. I liked the characterisation of fitness effects in a way that makes influenza and SARS-CoV-2 directly comparable. This is a neat new result.

I thought the retrospective regression against numbers of variants was a simple and interesting analysis. There is some precedent for examining correlates of fitness in key viral sites, for example in Bush et al., 1999 and Luksza and Lassig 2014: This carried on a useful tradition.

This paper is clearly written and well-constructed. The figures are well produced. There is an absence of hype in presenting results in a simple and straightforward manner.

After posing the question, I came to the conclusion that the comparison between S1 mutations and EVEscape was a fair one. The question here is that the details of the relationship between S1 mutations and fitness is derived in retrospect, from a large quantity of data, whereas PLM-based scores are advertised as having utility on the basis of only a single genome sequence. However, comparison was made with a fine-tuned model, which in some ways levels the playing field. Maybe further discussion could have been added around this point?

Weaknesses:

(1) I wasn't clear about whether the comparison with Fisher's fundamental theorem was a useful check on the analysis, or simply an inevitable consequence of any estimate of fitness scores. The fitness of a strain is measured in relative terms, as the difference in a fitness score relative to the mean of that score for the whole population. More rapid adaptation occurs

when the relative difference to the mean is large, which may necessarily correlate with an increased overall variance in the fitness of strains.

(2) I would have valued more discussion on the potential implications of the findings of this manuscript. Two questions I was left with:

a) Thinking about application to a potential future emergent pathogen, to what extent is prior knowledge being leveraged in these methods? While a comparison is made of fitness gain versus statistics of multiple genes, in one sense we 'know' that the more accessible parts of the viral glycoprotein are going to be of higher relevance for viral fitness and evolution, simply from an understanding of the necessary geometry of antibody-viral interactions. Similar knowledge is leveraged in the Luksza and Lassig 2014 study: Characterising fitness by a number of epitope substitutions in that case rests on biological knowledge of which are the important epitope sites. The EVE family of methods use a similar approach where factors such as residue accessibility are combined with outputs from protein language models. But does this work have implications for how we evaluate emerging variants of existing viruses, notably SARS-CoV-2 or influenza? The H3N2/K substrain of influenza makes me want to ask this question: In that case a large number of non-synonymous changes in HA was not accompanied by a noticeable increase in the severity of seasonal influenza.

(b) What is the implication of this work for the use of PLM-based models for understanding virus evolution? One analysis that was not done is evaluating the performance of combined models, encompassing both mutation counts and PLM data. Models such as EVEscape already utilise multiple metrics alongside PLMs. Do PLMs contribute addition knowledge to a 'number of mutations on the outside of the glycoprotein' count?

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Reviewer #2 (Public review):

Summary:

The author uses multinomial logistic regression to assign fitness values to SARS-CoV-2 and Influenza H3N2 clades. The mean population fitness, referred to as the cumulative fitness flux, is tracked throughout evolution to describe the rate of adaptation. The author shows a clear decrease in the rate of adaptation for SARS-CoV-2 after mid 2022, coinciding with the emergence of the Omicron clades; the rate of adaptation of H3N2 is much slower, but inferred to be constant in time. With this analysis, the author confirms Fisher's fundamental theorem of natural selection. Using a fine-grained clade definition scheme, the author attributes changes in fitness to substitutions in different genomic regions, highlighting the importance of the spike and, in particular, the RBD region of spike. A fitness model based on a multinomial regression is shown to perform better than EvEscape and protein-language-model semanticity to predict changes in fitness.

Strengths:

The author provides an interactive reading experience on the *github.io* hosted website, which has additional interactivity with figures and offers simple manipulation of what is shown. The methods are well explained and are easy to follow. The conclusions of the study-the decrease in the rate of adaptation of SARS-CoV-2, mutations in Spike S1 as the main driver of evolution, and the Spike S1 mutational count as predictor of fitness change - are clear to follow.

Weaknesses:

(1) The main conclusions that (i) the rate of adaptation of SARS-CoV-2 has decreased after Jun 2022 and is lower than in Influenza H3N2 and (ii) that mutations in Spike S1 are the main

drivers of evolution are not novel.

(2) At the core, the multinomial logistic regression assumes a time-independent fitness value for each viral clade, while the clade fitness is not constant and depends on the population immunity, which will change over time. As a result, the fitness increase can be wrongly interpreted: The author observes a doubling of the mean fitness every 6 months from Jan 2021 to Jun 2022, and every 2.4 years from July 2022 to Dec 2025. We would then expect that the mean fitness in Dec 2025, $\overline{f}_t$, has increased more than 2 to the power 4, as compared to the mean fitness in Jan 2021, $\overline{f}_0$ ($\overline{f}_t > 2^4\ \overline{f}_0$). Is the mean viral fitness in Dec 2025 more than 16 times higher than in Jan 2021? Fitness differences between viral populations at different times cannot be meaningfully compared because the fitness landscape is time-dependent. The increase in fitness, without accounting for the change in population immunity, is in part an artifact of the multinomial regression.

(3) The multinomial regression treats the fitness landscape on which the virus evolves as static. However the fitness landscape is actually time-dependent, as population immunity changes. For a static fitness landscape, the cumulative fitness flux is the mean population fitness. However, for a dynamic fitness landscape, this is no longer the case: the (mean population) fitness changes both because of adaptation, and because changes of changes in the fitness landscape, i.e., an increase in population immunity will decrease fitness non-uniformly, also affecting individual selection coefficients s_i(t). This leads to cases where the cumulative fitness flux between t_0 and t_n can be positive, but the fitness remains the same. This study neglects the time-dependence of the fitness landscape and therefore incorrectly calculate the fitness flux, leading to an overestimation of the rate of adaptation.

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Reviewer #3 (Public review):

Summary:

Bedford considers how the fitness of SARS-CoV-2 and influenza H3N2 variants have changed over time. The fitness of viral variants are estimated from trajectories of variant frequencies using multinomial logistic regression (MLR). The MLR approach was previously developed and is well-validated. Changes in mean population fitness (i.e. fitness flux) is then derived from changes in the fitness of circulating variants through time.

Strengths:

The manuscript makes some noteworthy observations. For one, it is reassuring to know that fitness variation well predicts the rate of fitness change in viral populations, as expected under Fisher's fundamental theorem. Similarly, it is interesting that simple counts of mutations perform as well as much more sophisticated protein language models in predicting variant fitness.

Weaknesses:

My two main concerns both relate to the interpretation of fitness flux and how confident we should be that it is quantifying changes in absolute fitness as opposed to environmentally-dependent changes in fitness through time.

(1) Interpretation of fitness flux:

I'm not entirely convinced that mean population fitness can be meaningfully compared through time, at least not as a measure of fitness change on an absolute scale. Here, mean population fitness is defined as the weighted sum of the relative fitness of each variant, so changes in mean pop fitness necessarily reflect changes in relative fitness. To obtain a fitness

value of each variant that is comparable across time, a "scaffolding approach" is used to estimate window-specific effect and the global fitness (f_i) of each variant. The major assumption here is that by accounting for the window-specific effects, the global fitness of each variant becomes directly comparable. This assumption seems somewhat suspect. It seems impossible to know whether the window-specific effect is capturing the full environmentally-dependent component of fitness in a given window. There could also be a high degree of temporal autocorrelation in window-specific effects between neighboring windows due to similarities in environment that would decay over time.

To put this in another way, say we directly competed a 2020 SARS-CoV-2 variant with a 2025 variant in the distant future after all host-immunity to these strains had waned. Would we really be confident in estimating the fitness advantage of the 2025 variant over the 2020 variant? Or would this estimates of relative strain fitness still reflect the environments in which they were measured in, including the NPIs that were in effect in 2020 and the complex immune structure of the host population in 2025?

(2) Inability to separate different components of fitness:

Earlier work, like Meijers et al. for SARS-Cov-2 (Cell, 2023), decomposed overall changes in fitness into different components of fitness such as changes in intrinsic fitness (transmission potential) and immune escape. While the current model is simpler, in some ways this earlier work was more elegant and more powerful in its ability to explain what components of fitness drove the overall change in population-level fitness.

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